DELPHI: which laws of biology hold up

Distributed Epistemic Layer for Probabilistic Hypothesis Inference. The goal: the fundamental theorems of biology, general laws stated precisely enough to test. We pull candidate laws out of the literature as claims, prove what can be proved and test the rest on real organisms.

extract
candidate laws from books and papers, with the quote

formalize
state it in Lean 4 where it is mathematics

simulate
reproduce the model, audit it blind

test
pre-register pass/fail, then fetch the data

publish
every step a signed nanopublication

1,761
claims, each with a verbatim quote from its source

87
theorems: 71 machine-checked in Lean 4, 16 proved in the literature

109
MorphoEval audits of model reproductions, linked to claims

161
claims tested against data, 82 of them on measured organisms

880
live as signed nanopubs on the production network

Sources so far: Darwin's Origin of Species, Thompson's On Growth and Form, Wolfram's A New Kind of Science, sixteen Levin papers, Aristotle's biology and five mathematical biologists. 782 animal taxa carry their own datasets.

Any agent can query it, no install: MCP server at delphisci-mcp.fly.dev/mcp (claims, their math, rankings, test verdicts, organisms).

A proof shows a law follows from its assumptions, not that organisms satisfy them. That is what the data tests are for.

Five people, a few hundred agents

Thirty hours at Science Claw Hack @ MIT: 101 local agent chats, 262 subagents and 19 cloud agents working with five people, plus the outside agents that challenged our tests.



Joel Dietz @fractastical
lead, integrator, publisher; signs every production nanopub

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form simulator and queryable open problems

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presentation layer: claim graph site, demo, slides

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nanopubs; linking claims to INDRA, GBIF and Wikidata

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genotype–phenotype links, MCP testing, agent comparison

and the agents

Claude builds · **GPT-5.6 Sol, Gemini 4 Argon, Grok 4.7** assess blind · **19 cloud agents** test **Delfie** (our agent) on **Moltbook,** The Colony and Discord, and as **delfie-claude** in the SentiLab room

Moltbook agents who challenged our cell-geometry test: **moltbook-vina, moltbook-hivefound, moltbook-nanoswarm, doctor_memory**

Also in the collaboration log: **InquilineKea** (SentiLab), **Fittest MCP**

What we solved: from books to verdicts

We built the machinery that takes a claim from a page to a pre-registered test on real data and a published verdict, and ran it. No open problem is solved yet; 161 claims have a verdict.

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claims from six sources, each with a verbatim quote

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VERDICTS ON THE 161 TESTS



Real refutations

Darwin's island biogeography (2 refuted, 2 mixed) and Thompson's heart and fish-shape claims were refuted on public data. Wolfram's randomness batteries and Levin's keratocyte and planarian-head claims held.

It audits itself

A blind assessor from another provider checks that each test measures what its claim says: 28 of 161 checked so far, 5 judged not representative and set aside. A segmenter-bias control cut one Thompson result from 0.60 to 0.35.

The collective beats one agent

Same model and budget, 20 claims, first of two experiments: one agent verified 13 datasets, isolated workers 17, a centralized swarm 19, a decentralized one 18, with no fabricated identifier in any final plan. Every multi-agent arm beat one agent in both.

Not solved: any of the 27 open problems. A verdict on one dataset is evidence, not proof.

What we can do next

The problem graph already holds 27 hard open problems, each linked to the claims it bears on, with approaches proposed blind by agents from four model providers. None is solved yet.

7 Prove it

Tier A: a proof or an exhaustive computation the swarm could finish. Rule 110 universality in Lean, the scope of Hamilton's rule, phyllotaxis attractors.

13 Test it

Tier B: open empirical questions a pre-registered test on public data can settle. Metabolic scaling, species–area, Murray's law, Lewontin's paradox.

7 Attack it

Tier C: hard and open, where agents have an unusual angle. The bioelectric code, morphogenesis–cognition gene overlap through INDRA, memory in gene networks.

Close the evidence gap

1,524 of 1,761 claims are neither proved nor tested. 9 pre-registered tests wait for data, and 133 of the 161 finished tests have not yet been through the faithfulness check.

Replicate across organisms

A law that holds in one organism is not fundamental. 82 of 161 tests used measured organism data; 782 taxa and 1,372 dataset records are ready to run each law across species.

Open to other teams

The hosted MCP server, agent forum and Frontier node (1,190 claims mirrored) already let outside agents challenge our tests, as the Moltbook agents did. Every new test lands as a nanopub.